Regularization Near the Interpolation Threshold: An Empirical Comparison of Minimum-Norm Regression, Ridge, and PCR

Yunjie Guo Hanqing Yang

Abstract

We compare minimum-Euclidean-norm least squares, cross-validated ridge regression, and principal components regression (PCR) as the training sample size crosses the feature dimension. Two fixed-holdout evaluations use YearPredictionMSD (90 predictors; an artist-disjoint holdout) and Superconductivity (81 predictors; an exact-feature-grouped holdout). Each has 100 training-subset permutations, 15 sample sizes, and source-only tuning for ridge and PCR. Ridge has the lowest mean holdout mean squared error (MSE) at every size on both benchmarks, and all 30 pointwise paired intervals for PCR minus ridge exclude zero. Because features are centered on the training data, the interpolation boundary lies at $n = p + 1$; minimum-norm loss rises sharply on both sides of it. On Superconductivity, a single repetition contributes 99.92% of total minimum-norm loss at $n = 73$, so the mean and median curves peak at different sizes. Post-hoc spectral diagnostics trace this tail to a few repetitions whose smallest retained singular value is only 9 to about 5,300 times the pseudoinverse cutoff. The same tail makes eight pointwise mean-difference intervals include zero, even though ridge has the lower loss in every repetition at those sizes. On YearPredictionMSD, ridge improves on a training-mean predictor by only about 3% near the threshold. Across these two holdouts, regularization mitigates severe near-threshold risk. The experiments neither establish a universal ranking nor identify a causal effect of distribution shift.

Keywords: minimum-norm least squares; ridge regression; principal components regression; interpolation threshold; heavy-tailed prediction error; ill-conditioning.

1 Introduction

The behavior of least squares changes markedly when the number of training observations n approaches the number of predictors p. In ordinary full-rank regression, sufficiently many observations constrain the fitted coefficients. Near rank deficiency, weak singular directions amplify perturbations in the response and can yield unstable out-of-sample predictions. When $n < p$, the minimum-Euclidean-norm solution is a common, well-defined way to select among least-squares solutions, but its predictive behavior depends on the geometry of the design matrix, the signal, and the evaluation distribution [Hastie et al., 2022]. The interpolation boundary and the possibility of nonmonotone risk curves have attracted renewed interest through work on double descent [Belkin et al., 2019]. A peak is not inevitable for every dataset and estimator, and empirical curves require careful interpretation.

Two classical regularization strategies act on this geometry differently. Ridge regression continuously attenuates components associated with small singular values [Hoerl and Kennard, 1970]. Principal components regression (PCR) truncates the singular expansion to a selected number of

leading components [Massy, 1965; Hastie et al., 2009]. High-dimensional analyses show how spectral structure and the aspect ratio affect predictive risk under explicit distributional assumptions [Dobriban and Wager, 2018; Hastie et al., 2022]. These results motivate, but do not predetermine, the ranking of source-tuned estimators on finite real-world holdouts.

Closest to our question, Nakkiran et al. [2021] show that optimally tuned ridge regularization can remove the double-descent peak of linear regression, both in theory under isotropic designs and in experiments. Advani et al. [2020] analyze the same phenomenon through the dynamics of generalization error. Our contribution is complementary rather than new in mechanism. We examine whether source-only cross-validation, rather than oracle tuning, delivers this protection on two real benchmarks with fixed external holdouts. We also document how heavy-tailed repetition-level losses affect the summaries and paired inference that such comparisons usually report.

The study asks a deliberately empirical question: how do minimum-norm regression, source-cross-validated ridge, and source-cross-validated PCR differ in fixed-holdout prediction risk and training-subset stability as n passes through the interpolation boundary? The original research plan emphasized YearPredictionMSD's artist-disjoint holdout and asked whether qualitative rankings would recur on a Superconductivity control. The executed control uses an exact-feature-grouped split, not the IID-style row split originally described. The two experiments are therefore separate benchmark protocols, not a design that isolates a distribution-shift effect.

We report four connected observations. First, ridge attains the lowest mean holdout MSE at all 15 sizes on each dataset. Second, minimum-norm deteriorates severely near the threshold, but its mean and its typical repetition can tell very different stories: on Superconductivity, an extreme right tail moves the mean maximum well below the threshold. Third, such tails can make pointwise Student-t intervals for the mean paired difference cross zero even when ridge wins every observed repetition. Fourth, post-hoc spectral diagnostics link the extreme tail to retained singular values barely above the pseudoinverse cutoff. We do not propose a new estimator or claim a new theory.

2 Methods

2.1 Estimators and spectral filters

For a training prefix of size n, let $Z \in \mathbb{R}^{n\times p}$ denote the centered, scaled predictor matrix and let $z = y - \bar{y}_{\text{train}}\mathbf{1}$ denote the centered response. With the economy singular value decomposition

$$Z = U\,\mathrm{diag}(s_1,\dots,s_d)V^\top, \qquad s_1 \geq \cdots \geq s_d \geq 0, \qquad d = \min(n,p), \tag{1}$$

we compare the following coefficients, all expressed in the common transformed feature space.

Minimum-norm least squares. The implementation uses the float64 SVD pseudoinverse

$$\hat{\beta}_{\text{MN}} = \sum_{j:\, s_j > \tau} \frac{u_j^\top z}{s_j}\, v_j, \qquad \tau = \epsilon_{64}\max(n,p)\, s_1, \tag{2}$$

where ϵ_{64} is the float64 machine epsilon. Components at or below τ are discarded. Above the cutoff, arbitrarily small retained s_j still enter through $1/s_j$. Because features are centered, the training intercept is restored after prediction rather than penalized.

Centering and the interpolation threshold. Because the columns of Z are centered on the training prefix, $Z^\top\mathbf{1} = 0$. If the prefix contains $m \leq n$ distinct feature vectors, then $\mathrm{rank}(Z) \leq$

Table 1: Two fixed-holdout benchmark protocols. The internal tuning folds are row-level and are distinct from Superconductivity's grouped outer split.

Characteristic	YearPredictionMSD	Superconductivity
Rows in complete dataset	515,345	21,263
Predictors p	90	81
Source-pool rows	463,715	16,871
Fixed-holdout rows	51,630	4,392
Holdout construction	Artist-disjoint	Exact-feature grouped
Repetitions	100	100
Training sample sizes	15	15
Methods	Minimum-norm; ridge; PCR	
Primary risk measure	Original-scale unweighted holdout MSE	

$\min(m-1, p)$. With distinct rows ($m = n$), the minimum-norm fit interpolates the training responses whenever $n - 1 \le p$, so the effective interpolation threshold is $n = p + 1$ rather than $n = p$. For $n \le p$, the singular value associated with the direction $\mathbf{1}/\sqrt{n}$ is zero in exact arithmetic and is removed by the cutoff τ. This truncation is structural and does not by itself indicate numerical difficulty. Repeated feature vectors, which occur in Superconductivity training prefixes, add further exact null directions. Neither grid contains $n = p + 1$ (91 and 82, respectively); the nearest grid sizes are 90 and 92 for YearPredictionMSD and 81 and 83 for Superconductivity.

Ridge regression. For a strictly positive penalty λ, the objective is

$$\hat{\beta}_\lambda = \arg\min_\beta \left\{ \frac{1}{n}\|z - Z\beta\|_2^2 + \lambda\|\beta\|_2^2 \right\} = \sum_{j=1}^{d} \frac{s_j}{s_j^2 + n\lambda} (u_j^\top z)\, v_j. \tag{3}$$

Ridge therefore shrinks small-singular-value directions smoothly rather than applying the minimum-norm hard cutoff. We tune λ over 25 logarithmically spaced values from 10^{-8} to 10^4, inclusive, using only the sampled training prefix. This scaling of λ corresponds to the $1/n$ residual objective, not an unnormalized sum of squares.

Principal components regression. For a selected component fraction q and numerical rank $r = \#\{j : s_j > \tau\}$, the implementation retains

$$k(q, r) = \min\{r, \max\{1, \lceil qr \rceil\}\}, \qquad \hat{\beta}_{\mathrm{PCR},q} = \sum_{j=1}^{k(q,r)} \frac{u_j^\top z}{s_j}\, v_j. \tag{4}$$

The 12 candidate fractions are $\{1/20, 1/10, 1/5, 3/10, 2/5, 1/2, 3/5, 7/10, 4/5, 9/10, 19/20, 1\}$. The integer k is recomputed from each fold's own numerical rank; a selected fraction, rather than an integer component count, is refitted on the full training prefix. A rank-zero PCR fit is treated as an error, not silently replaced by an intercept-only fit. These filters distinguish smooth ridge shrinkage from PCR's component truncation and from the unpenalized inversion of all retained minimum-norm directions.

2.2 Data and fixed holdouts

YearPredictionMSD contains 515,345 songs represented by 90 audio features and their release years [Bertin-Mahieux, 2011]. We use the repository's prescribed separation: the first 463,715 rows form

the source pool and the final 51,630 rows form the fixed holdout. UCI describes this arrangement as preventing songs from the same artist from appearing on both sides of the split. We therefore call it an artist-disjoint evaluation, without assuming that it isolates covariate shift or any other single statistical mechanism.

Superconductivity contains 21,263 rows, 81 material-derived predictors, and the continuous target `critical_temp` [Hamidieh, 2018b,a]. The outer split groups rows by exact equality of their entire 81-dimensional feature vector, without using the target. A fixed random permutation assigns $\lfloor 0.8G \rfloor$ of the $G = 15{,}170$ distinct groups to the training side and the remainder to the holdout. This yields 12,136 training groups (16,871 rows) and 3,034 holdout groups (4,392 rows), with no identical feature group crossing the boundary. It is a duplicate-aware grouped random holdout, not a pure independent row split. Because each group stays on one side, training prefixes can contain repeated feature vectors. Table 1 summarizes both protocols.

2.3 Sampling, preprocessing, and tuning

Each benchmark has a prespecified list of 100 distinct pseudorandom seeds. For repetition r, one seed permutes the entire ordered source-pool row vector; the training sample at size n is its nested prefix of length n. All three estimators use exactly that prefix. YearPredictionMSD uses $n \in \{45, 68, 81, 86, 88, 90, 92, 95, 99, 113, 135, 180, 270, 450, 900\}$ and Superconductivity uses $n \in \{41, 61, 73, 77, 79, 81, 83, 85, 89, 101, 122, 162, 243, 405, 810\}$. The two grids deliberately straddle $p = 90$ and $p = 81$, respectively, and reach $n/p = 10$.

Preprocessing is fitted exclusively on a training prefix, or on the training portion of an inner cross-validation fold. Each feature is centered and divided by its training-population standard deviation (`ddof=0`). Zero-variance columns are retained with scale set to one. The response is centered at the training mean, which is added back to every validation and holdout prediction. Neither validation nor holdout rows contribute preprocessing statistics.

Ridge and PCR use the same five source-only folds and minimize the unweighted mean of five fold-level MSEs. Fold membership is assigned once to the maximum prefix length in each benchmark and restricted to smaller nested prefixes, so an existing position keeps its fold identity as the sample grows. Fold sizes are therefore unbalanced at small n; the smallest fold has five rows (YearPredictionMSD at $n = 45$ and Superconductivity at $n = 41$), and no fold is smaller. At exact score ties, the earliest candidate in the ascending grid wins; PCR fractions that yield the same integer component count remain distinct candidates. Minimum-norm is not tuned. A failed candidate is unscorable. If no required candidate succeeds, or a required final fit fails, the run aborts rather than replacing a repetition. All 100 repetitions were retained in both experiments.

For Superconductivity, outer grouping does not imply feature-disjoint inner folds. An independent feature-only audit found exact-feature cross-fold overlap in 807 of 1,500 repetition-size cells and 2,574 of 7,500 fold evaluations, involving 10,270 duplicate-feature validation exposures (maximum affected fraction in a fold: 0.20). At $n = 810$, every repetition and all 500 fold evaluations were affected, with 6,226 of 81,000 validation-row exposures. These folds were fixed in the protocol and were not changed after observing outcomes. We consider their interpretive consequences in Section 4.

2.4 Risk endpoint and paired uncertainty

Let H be the fixed holdout and $\hat{f}_{r,n,m}$ the fitted estimator $m \in \{\mathrm{MN}, \mathrm{Ridge}, \mathrm{PCR}\}$ on repetition r and size n. Its original-scale holdout risk is

$$L_{r,n,m} = \frac{1}{|H|} \sum_{i \in H} \left(y_i - \hat{f}_{r,n,m}(x_i)\right)^2, \qquad \bar{L}_{n,m} = \frac{1}{100} \sum_{r=0}^{99} L_{r,n,m}. \tag{5}$$

The prespecified primary endpoint is $\bar{L}_{n,m}$. Medians, quartiles, extremes, and prediction quantiles are secondary descriptive diagnostics. For within-repetition comparisons, $D_{r,n}^{a-b} = L_{r,n,a} - L_{r,n,b}$. With $R = 100$ complete repetitions, the prespecified uncertainty summaries are

$$\bar{D} \pm t_{0.975,99} \frac{s_D}{\sqrt{100}}, \qquad s_D^2 = \frac{1}{99} \sum_{r=0}^{99} (D_r - \bar{D})^2, \qquad t_{0.975,99} \approx 1.9842. \tag{6}$$

The specified signs are PCR minus ridge, minimum-norm minus ridge, and PCR minus minimum-norm. These are pointwise, not simultaneous, intervals describing variation over the specified source-pool permutations conditional on each fixed holdout. They are not a target-population uncertainty model. For cross-dataset visual comparison we use n/p and the ratio of mean risks, $Q_{n,m} = \bar{L}_{n,m}/\bar{L}_{n,\mathrm{Ridge}}$; this is not the mean of repetition-level ratios.

2.5 Post-hoc descriptive diagnostics

After both confirmatory runs were complete, we ran a separate set of descriptive diagnostics. They were not prespecified, did not rerun or modify the confirmatory pipeline, and do not replace any primary result. They are as follows.

1. *Regeneration.* Training prefixes were rebuilt from the stored seeds, and all three estimators were refitted with the archived selected hyperparameters at eight cells, including the largest-loss cells on each benchmark. All 24 recomputed holdout MSEs matched the archived values exactly (maximum relative error 0).

2. *Repetition-level spectra.* For every repetition at Superconductivity sizes $n = 41$–89 (nine grid sizes) and YearPredictionMSD sizes $n = 81$–99 (seven grid sizes), we recorded the singular values of Z, the cutoff τ, the number of truncated values, the smallest retained value $s_{\min}$, the distribution of $\|\hat{\beta}_{\mathrm{MN}}\|_2^2$ across directions, and the number of distinct feature rows.

3. *Source-pool spectrum.* Eigenvalues of the correlation matrix of each full source pool.

4. *Null baseline.* The training-prefix mean as a predictor, with the paired ridge-minus-null interval of Eq. (6).

5. *Selected hyperparameters* and inner-fold sizes, read from the archived records.

6. *Exploratory paired summaries* (exact sign tests and paired $\log_{10}$-MSE differences), reported only in Appendix C.

Extended-precision recomputation was planned but not run, because no extended-precision arithmetic was available in the analysis environment.

Table 2: Mean fixed-holdout MSE at selected sample sizes. Values are arithmetic means over all 100 training-subset repetitions, not medians. "Mean only" predicts the training-prefix mean (post-hoc reference). Full grids appear in Appendix A.

Dataset	n (n/p)	Minimum-norm	Ridge	PCR	Mean only
Year	45 (0.50)	422.27	121.73	133.25	121.34
Year	90 (1.00)	48,972.31	115.78	123.20	119.21
Year	900 (10.00)	104.68	99.50	105.48	117.92
Super	41 (0.51)	8.837×10^9	575.29	689.78	1176.02
Super	73 (0.90)	3.238×10^{19}	509.82	610.22	1160.67
Super	81 (1.00)	2.698×10^{17}	491.10	594.77	1158.07
Super	810 (10.00)	341.26	333.63	345.19	1146.60

Table 3: Pointwise 95% paired intervals excluding zero, out of 15 tested sizes. Each difference is the first estimator's MSE minus the second's. Counts are descriptive across multiple pointwise intervals, not familywise-adjusted inference.

Paired difference	YearPredictionMSD	Superconductivity
PCR − ridge	15/15 (all positive)	15/15 (all positive)
Minimum-norm − ridge	15/15 (all positive)	7/15 (all positive)
PCR − minimum-norm	15/15 (14 negative, 1 positive)	7/15 (6 negative, 1 positive)

The positive PCR − minimum-norm interval occurs at the largest size, $n/p = 10$, in both benchmarks. On Superconductivity, the seven minimum-norm − ridge and PCR − minimum-norm intervals that exclude zero are at $n \in \{89, 101, 122, 162, 243, 405, 810\}$.

3 Results

3.1 Ridge ranks first in mean risk on both holdouts

Across all 15 sizes in each benchmark, source-CV ridge has the lowest mean holdout MSE. PCR is next near the interpolation boundary, and minimum-norm is substantially worse there. At the largest size, $n/p = 10$, minimum-norm moves slightly ahead of PCR but remains behind ridge. Table 2 gives selected risk levels, and Figure 1 shows the complete 15-point curves. Because release year and critical temperature have different scales, the two datasets' absolute MSEs cannot be compared numerically.

For PCR versus ridge, all 30 pointwise 95% paired intervals exclude zero, with PCR having the larger mean MSE. For minimum-norm versus ridge, all 15 YearPredictionMSD intervals exclude zero, but only 7 of 15 Superconductivity intervals do. This is not a contradiction of the mean ranking: the heavy-tailed variability of minimum-norm inflates its standard errors (Section 3.4). Table 3 summarizes these comparisons.

Reference: the training-mean predictor. Predicting the training-prefix mean gives a mean holdout MSE of 117.9–121.3 on YearPredictionMSD and 1146.6–1176.0 on Superconductivity, close to the holdout target variances of 117.76 and 1143.32. On Superconductivity, ridge improves on this baseline by 51% at $n = 41$ and 71% at $n = 810$. On YearPredictionMSD, the gain is small. At $n = 45$, ridge's mean MSE is 0.33% higher than the baseline's, and the paired interval for ridge minus baseline, $[-1.20, 1.98]$, includes zero. From $n = 68$ onward every such interval excludes zero, but the improvement is only 2.4–3.1% across $n = 81$–99, rising to 15.6% at $n = 900$. On YearPredictionMSD, ridge's advantage near the threshold therefore lies mostly in avoiding the

minimum-norm failure, not in substantial predictive gains. Consistent with this, ridge selected the largest grid penalty, $\lambda = 10^4$, in 135 of 1,500 YearPredictionMSD cells, concentrated at the smallest sizes (Figure 7).

3.2 A sharp increase in minimum-norm risk around the threshold

On YearPredictionMSD ($p = 90$), minimum-norm's mean MSE rises from 422.27 at $n = 45$ to 48,972.31 at $n = 90$, about 423 times ridge's mean of 115.78 at the same size. The two grid sizes bracketing the effective threshold $n = p + 1 = 91$ have nearly identical means (48,972.31 at $n = 90$ and 48,728.59 at $n = 92$). Among grid sizes, the median is also largest at $n = 90$ (15,483.77). The data therefore locate the peak between these two sizes, not at $n = p$ specifically. Ridge stays near 115 across the near-threshold grid, and PCR stays near 123–129 (Figure 1).

Superconductivity ($p = 81$) is more extreme and less symmetric. Its minimum-norm mean reaches 3.238×10^{19} at $n = 73$ ($n/p \approx 0.90$), well below the effective threshold $n = p + 1 = 82$. At $n = 81$ the mean is still 2.698×10^{17}, against ridge's 491.10. The median, however, is largest at $n = 81$, the grid size immediately below the threshold, at 2.409×10^7. This displacement of the mean peak from the median peak means the average-risk curve cannot be read as the performance of a typical repetition; Section 3.5 traces it to a few repetitions. The PCR-to-ridge ratio is comparatively modest, from 1.035 at $n = 810$ to 1.215 at $n = 85$ (Figure 2).

At $n/p = 10$, relative mean-risk gaps are much smaller. On YearPredictionMSD, ridge, minimum-norm, and PCR have means of 99.50, 104.68, and 105.48; on Superconductivity they are 333.63, 341.26, and 345.19. Ridge remains first in both, and its paired advantages at these sizes have pointwise intervals that exclude zero. This narrowing is an observed finite-sample pattern, not evidence of asymptotic equality.

3.3 Typical repetitions and catastrophic tails

To separate the center of the repetition distribution from its upper tail, Figure 3 displays $\log_{10}$ MSE boxplots for the three estimators around the threshold. On YearPredictionMSD at $n = 90$, the worst minimum-norm repetition contributes 25.1% of total summed MSE, and the median itself is also far above ridge. The peak therefore reflects deterioration in both typical and extreme outcomes.

On Superconductivity the effect is much stronger. At $n = 73$, the median minimum-norm MSE is 1.396×10^6, the first quartile is 3.093×10^5, and the mean is 3.238×10^{19}. One repetition (ID 5, seed 200000046) has MSE 3.235×10^{21} and supplies 99.92% of the sum of all 100 losses. Its maximum absolute prediction is 3.597×10^{11}, and 97.7% of its holdout predictions exceed 10,000 in absolute value. In the same repetition and at the same size, ridge has MSE 463.66 and maximum absolute prediction 112.2, and PCR has MSE 631.31 and maximum absolute prediction 135.3. The disparity is present in the predictions themselves, not merely in the aggregation of losses.

This does not make the minimum-norm median competitive: at $n = 73$ its lower quartile already exceeds ridge's mean by about 600-fold. In 48 of 100 repetitions at $n = 73$ and 68 of 100 at $n = 81$, minimum-norm produces at least one prediction whose absolute value exceeds 10,000. By $n = 810$, no such repetition remains, and the minimum-norm mean and median are 341.26 and 336.77. All repetitions enter the primary mean; outlier labels and medians are descriptive diagnostics, not a procedure for removing unfavorable observations.

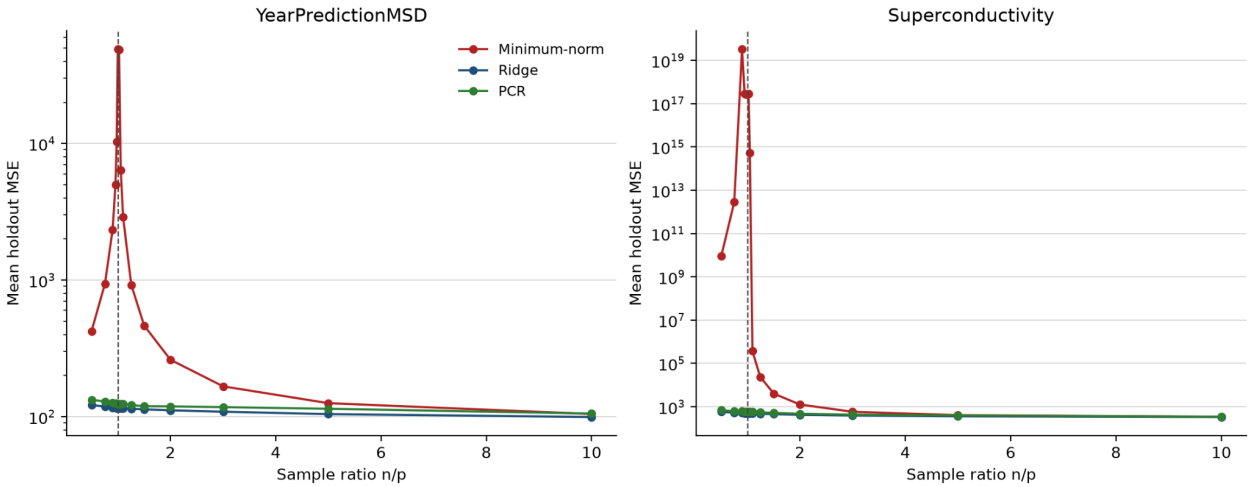


Figure 1: Original-scale mean holdout MSE against normalized sample size n/p on YearPredictionMSD (left) and Superconductivity (right). The vertical axes are logarithmic and dataset-specific. The dashed line marks $n/p = 1$; the effective interpolation threshold under centering is $n = p + 1$, just to its right. Every plotted point is an observed grid size; no interpolation of results was performed.

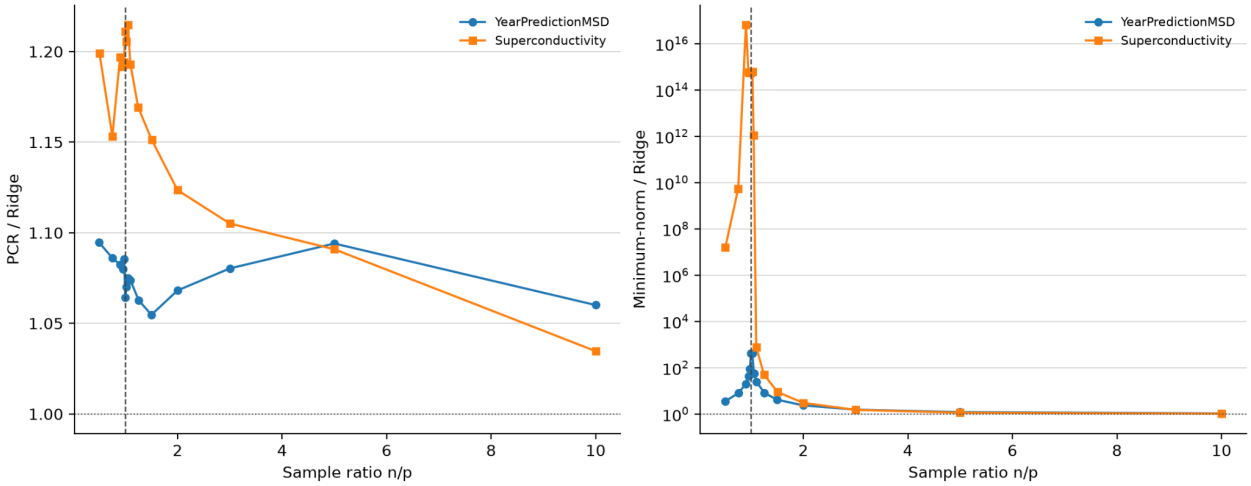


Figure 2: Mean-risk ratios relative to ridge on each fixed holdout: PCR/ridge (left) and minimum-norm/ridge (right). Ratios are computed from the corresponding means, not by averaging per-repetition ratios. Minimum-norm requires a logarithmic scale because of catastrophic Superconductivity losses. The dotted horizontal line is the ridge reference of one; the dashed vertical line is $n/p = 1$.

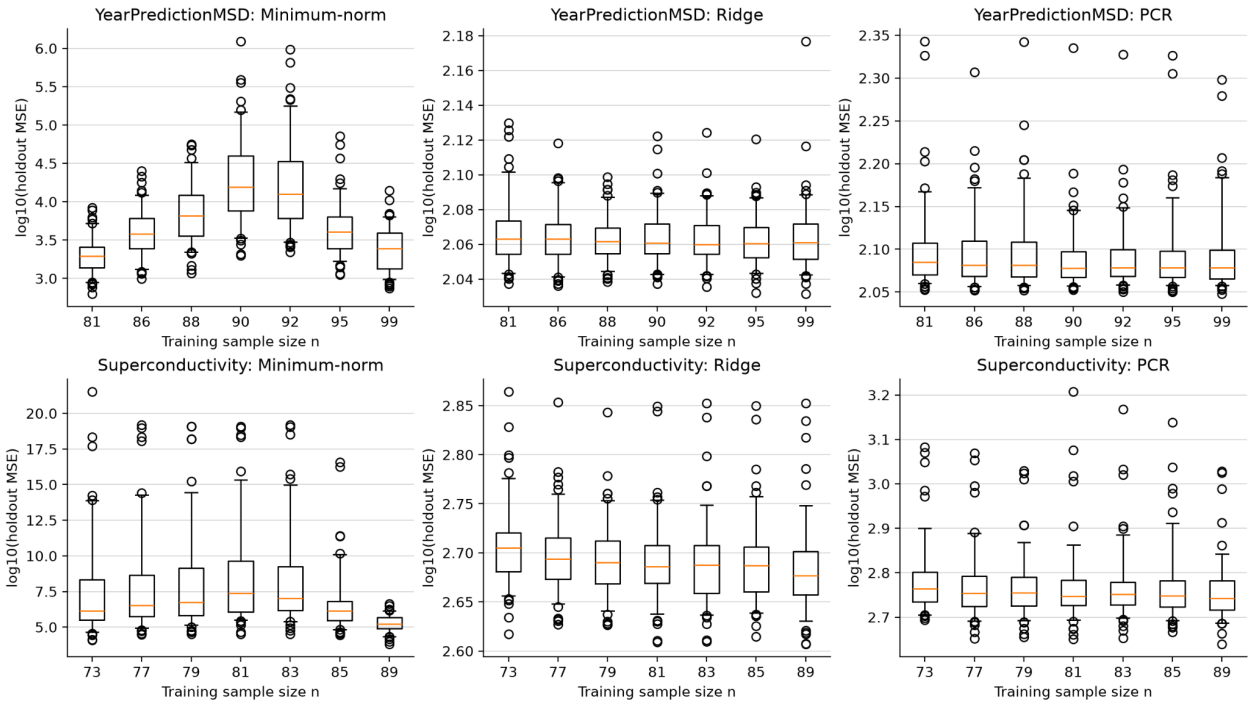


Figure 3: Repetition-level $\log_{10}$ holdout MSE around the interpolation threshold, by dataset and estimator. Boxes summarize the center of the 100-repetition distribution; whiskers depict the 5th and 95th percentiles, and values beyond them are shown individually. No repetition is trimmed or winsorized. Note the much wider vertical scale for Superconductivity minimum-norm than for the other methods.

3.4 Paired intervals under extreme right tails

Figure 4 presents all prespecified paired contrasts. On YearPredictionMSD at $n = 90$, minimum-norm minus ridge has mean difference 48,856.53 and pointwise 95% interval [22,299.94, 75,413.11]. At $n = 900$, the mean difference is 5.18 with interval $[4.79, 5.58]$. PCR minus ridge at those sizes has intervals $[5.21, 9.63]$ and $[5.53, 6.43]$.

On Superconductivity, minimum-norm-minus-ridge intervals include zero at eight sizes: $n = 41$, 61, 73, 77, 79, 81, 83, and 85. At $n = 73$, for example, the mean paired difference is 3.238×10^{19} and its pointwise interval is approximately $[-3.181, 9.657] \times 10^{19}$. Despite that interval's lower endpoint, ridge has the lower MSE in all 100 paired repetitions at this size, and at every tested size through $n = 243$. These facts coexist because a few enormous positive differences dominate the sample variance in the Student-t formula. The interval is an interval for a mean under the prespecified procedure; its inclusion of zero is not evidence of equal per-repetition performance. We do not replace this procedure with a post-hoc test. Exploratory sign tests and $\log_{10}$-scale paired intervals at these eight sizes, which are not prespecified, are reported separately in Appendix C.

Beyond the most extreme region, mean-difference intervals stabilize. At Superconductivity $n = 810$, minimum-norm minus ridge is 7.63 (pointwise 95% interval $[5.88, 9.38]$), and PCR minus ridge is 11.56 ($[9.68, 13.45]$). Minimum-norm first wins individual repetitions against ridge at $n = 405$ (3 of 100) and wins 17 of 100 at $n = 810$; on YearPredictionMSD it wins 2 of 100 at $n = 900$ and none at smaller sizes. The mean still favors ridge at every size. The PCR-minus-minimum-norm difference becomes positive at the largest size in both benchmarks, and both corresponding intervals exclude zero.

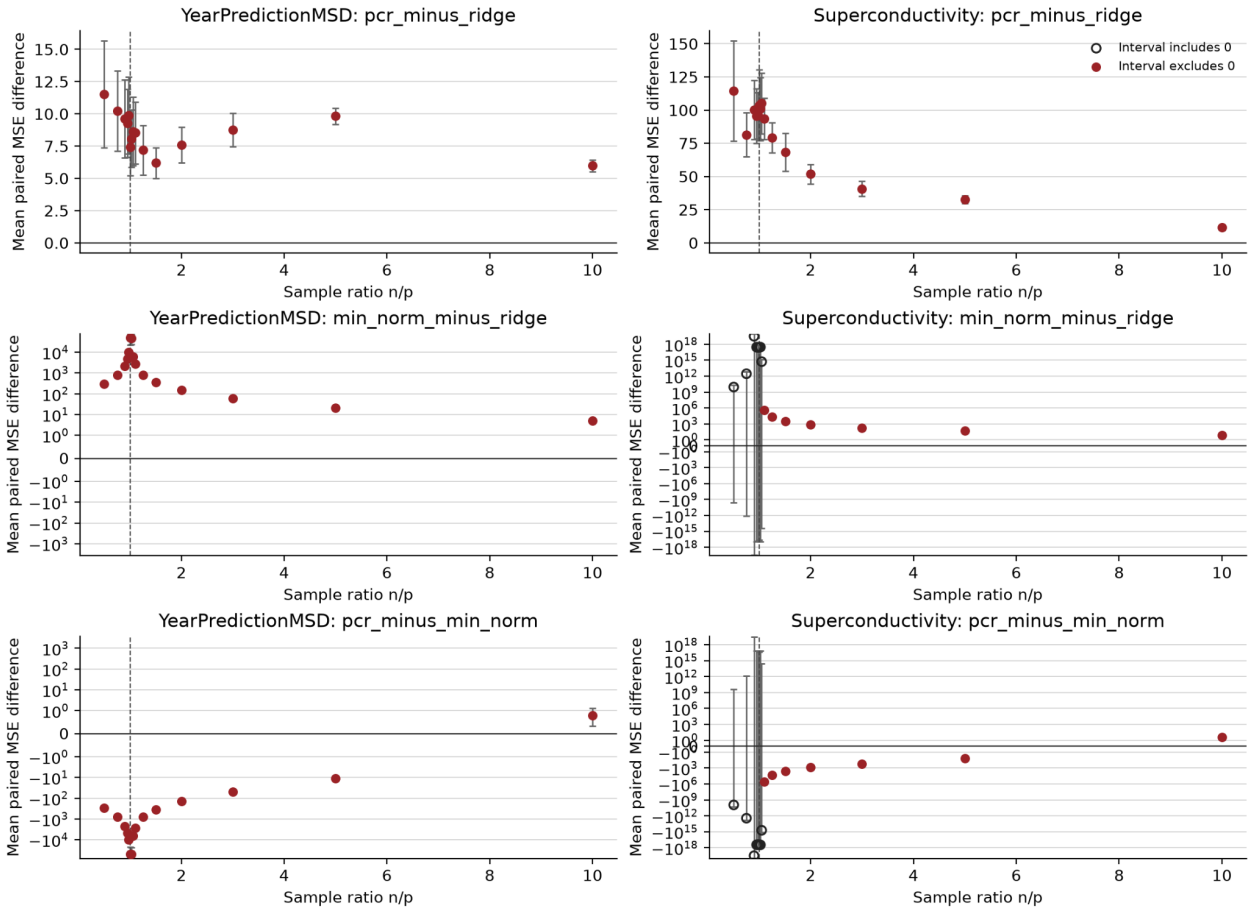


Figure 4: Means and prespecified pointwise 95% Student-t intervals for paired MSE differences across 100 training-subset repetitions, using the stated sign convention (panel titles). Filled markers denote intervals excluding zero; open markers denote intervals including zero. Signed-logarithmic display scales are used in the extreme minimum-norm panels so that negative interval endpoints remain visible. This display transformation does not alter the interval calculation.

3.5 Ill-conditioning near the threshold

The prediction arrays and loss records agree, and the largest Superconductivity errors occur in the same repetitions that produce extreme predictions; around $n = 73$ and $n = 81$, the repetition-level correlation between $\log_{10}$ MSE and $\log_{10}$ maximum absolute prediction is approximately 0.999. The post-hoc diagnostics of Section 2.5 examine where this amplification comes from. Regeneration reproduced every checked archived loss exactly, so the diagnostics below describe the same fits as the primary results.

Truncation warnings. The saved pseudoinverse warnings record that every repetition at Superconductivity $n \leq 81$ truncated at least one singular value. Section 2.1 shows this is expected. On YearPredictionMSD, every one of the 400 diagnosed cells with $n \leq p$ truncates exactly one singular value, whose left singular vector is aligned with $\mathbf{1}/\sqrt{n}$ ($|u^\top \mathbf{1}|/\sqrt{n} > 0.999$). On Superconductivity this simple pattern holds in 377 of 600 cells with $n \leq p$. Each of the remaining 223 cells has repeated feature rows and truncates additional exact null directions (up to four). At $n = 83$, 13 of 100 repetitions still truncate; all have repeated rows (typically 81 distinct rows among 83), and none has duplicated or constant columns. Truncation warnings therefore reflect centering and duplicate rows. They are not, by themselves, evidence of instability.

Smallest retained singular values. The informative quantity is the smallest singular value that is *not* truncated. Across repetitions, $\log_{10}$ MSE tracks $\log_{10}(1/s_{\min})$ closely on Superconductivity (Pearson correlation 0.92–0.98 for $n = 41$–85; 0.975 at $n = 73$) and more loosely on YearPredictionMSD (0.19–0.64, largest at $n = 92$; Figure 5). On Superconductivity, a few repetitions at each size from $n = 73$ to $n = 85$ have $s_{\min} < 10^{-10} s_1$, only 9 to about 5,300 times the cutoff τ. These two to six repetitions account for at least 99.987% of the summed minimum-norm loss at every one of these sizes (Table 7). The same repetition identities (notably 5, 57, 70, and 18) recur across nested sizes, as expected when the responsible rows stay in the prefix. For repetition 5 at $n = 73$, the prefix has 72 distinct rows among 73, and two singular values are truncated. The smallest retained value is $s_{\min} = 7.71 \times 10^{-12}$, against $\tau = 8.62 \times 10^{-13}$ and $s_1 = 47.94$, so $s_{\min}/s_1 \approx 1.6 \times 10^{-13}$. That single direction carries 99.99998% of $\|\hat{\beta}_{\mathrm{MN}}\|_2^2$. No YearPredictionMSD cell comes close: there, $s_{\min}/s_1 \geq 4.3 \times 10^{-5}$ and $s_{\min} \geq 2 \times 10^9\, \tau$ throughout.

Pool-level conditioning. The source pools differ sharply in spectral structure, even though both have full numerical rank and neither has duplicated or constant columns. In the Superconductivity correlation matrix, 62 of 81 eigenvalues lie below $10^{-2}\lambda_{\max}$ and 16 below $10^{-4}\lambda_{\max}$; the corresponding YearPredictionMSD counts are 4 of 90 and 0 (Figure 6). This near-degeneracy is consistent with Superconductivity's minimum-norm risk being already extreme far below the threshold (8.8×10^9 at $n = 41$), where the repetitions with near-cutoff singular values play no role.

Interpretation. The Superconductivity mean-risk peak at $n = 73$ reflects the entry of a near-cutoff direction into one repetition's prefix, not the location of the interpolation threshold. Descriptively, excluding the cells with $s_{\min} < 10^{-10} s_1$ would move the largest remaining mean to $n = 81$, matching the median. The primary mean includes every repetition and is unchanged. We cannot determine whether singular values of order $10^{-13} s_1$ represent exact linear dependence among the prefix rows or rounding residue of an exactly null direction, because extended-precision recomputation was not available. In either case, the catastrophic losses depend on directions that lie within a few orders of magnitude of the cutoff rule in Eq. (2). A larger cutoff would have discarded them; we

did not evaluate the resulting risk. Away from these cells, the amplification of the remaining small singular values is the ordinary ill-conditioning that ridge's filter in Eq. (3) is designed to control.

4 Discussion

The two evaluations give a coherent empirical picture. Under source-only selection and fixed external holdouts, ridge gives the lowest mean MSE at every training size, whereas minimum-norm becomes particularly fragile around the interpolation threshold $n = p + 1$. This ranking does not follow as a theorem from the experimental design, but it is consistent with the spectral interpretation of ridge: the filter $s/(s^2+n\lambda)$ in Eq. (3) limits the contribution of weak singular directions that enter Eq. (2) through division by s. PCR offers a different compromise: deleting the weaker components reduces risk near interpolation but can discard predictive directions that ridge retains with attenuated weights. The benchmarks show the realized trade-off under cross-validated settings, not a general preference valid for all covariance and signal structures [Hastie et al., 2009; Dobriban and Wager, 2018]. The finding that tuned ridge suppresses the peak agrees with Nakkiran et al. [2021]; here the tuning is by finite, source-only cross-validation rather than by an oracle.

The most distinctive empirical feature is not simply a large minimum-norm mean. On YearPredictionMSD, the center of the repetition distribution and its upper tail both degrade at the threshold. On Superconductivity, the center also deteriorates dramatically, but a few repetitions with singular values just above the pseudoinverse cutoff dominate the arithmetic mean so heavily that its observed peak occurs at $n = 73$ rather than next to $n = 82$. Presenting only Figure 1 would obscure this. The distributional display in Figure 3, the worst-repetition prediction magnitudes, and the spectral diagnostics show why the mean-risk curve should be read together with typical-repetition behavior. They also suggest that, for nearly collinear designs with repeated rows, the pseudoinverse tolerance is itself a consequential modeling choice.

A related statistical lesson concerns uncertainty for heavy-tailed paired losses. The prespecified interval uses the sample standard deviation of 100 differences. When one observation contributes nearly all of the sum, the variance is extremely large, and the nominal symmetric Student-t interval can extend below zero even when every observed difference is positive. Neither the positive sample mean nor the descriptive 100/100 win count substitutes for the other; they target different summaries. We keep the prespecified interval rather than introducing favorable post-hoc inference. In this regime, its approximate-coverage interpretation should be treated cautiously, because 100 observations may resolve a very heavy right tail poorly.

The cross-validation design interacts with the threshold. Inner training folds contain n minus one fold's rows, which is roughly $0.73n$–$0.87n$ at the sizes near p. For outer sizes up to about $1.1p$, every inner fit therefore lies on the underdetermined side of the threshold, even when the final fit does not. The selected penalties nevertheless show no abrupt change across the threshold. The median $\log_{10}\lambda$ moves from 0.75 at $n = 90$ to 0.50 at $n = 92$ on YearPredictionMSD, and stays at -0.50 at $n = 81$ and $n = 83$ on Superconductivity. Penalties at the grid edges are rare except for the 135 YearPredictionMSD cells at $\lambda = 10^4$ noted above, and one Superconductivity cell sits at $\lambda = 10^{-8}$ (Figure 7).

The external holdout designs provide useful but different forms of separation. YearPredictionMSD prevents cross-split artist overlap as described by UCI; Superconductivity prevents an exact 81-feature duplicate group from appearing on both sides. The original exploratory goal included comparison against an IID-style control, but the executed control is an exact-feature-grouped split. Since the datasets, targets, and holdout constructions differ simultaneously, their shared ranking is a replication across two evaluation settings, not identification of a distribution-shift effect. The

Superconductivity inner folds also retain known duplicate-feature overlap despite the protected outer split. Such overlap can influence source-only hyperparameter selection, which limits how broadly the tuned-estimator ranking can be interpreted, but it does not invalidate the recorded grouped-holdout losses.

Finally, the advantage of ridge over minimum-norm declines markedly between $n/p \approx 1$ and $n/p = 10$ in both studies. At the largest observed size, the difference is small in the original target units but still separated by the pointwise paired interval. We do not extrapolate this finite trend to infinite-sample equality. On YearPredictionMSD, all three estimators also remain close to the training-mean baseline until n is several times p, so the practical stakes of the comparison there are mainly about avoiding catastrophic failure.

5 Limitations

This study has two benchmarks and one fixed holdout per benchmark. Its empirical ranking is conditional on those data, the three estimator implementations, the tuning grids, and the chosen sample sizes. The 95% intervals are pointwise summaries across sampled training-subset permutations conditional on the fixed holdout, not simultaneous error bands or statements about a population of test observations. The nested samples share many training rows, and the loss distribution is exceptionally heavy-tailed for Superconductivity around the threshold, which limits the practical reliability of a normal-approximation-style reading of the reported Student-t intervals there. Because features are centered, the effective interpolation threshold is $n = p + 1$, which neither grid contains; the location of the peak is resolved only to the two bracketing grid sizes.

The YearPredictionMSD split and Superconductivity grouped split do not form a matched shift experiment. Within the Superconductivity source pool, exact-feature duplicates cross inner CV folds in the documented proportion, so those tuning folds are not feature-disjoint. The spectral, baseline, and hyperparameter diagnostics were defined after the primary results were known; they are descriptive and are not part of the prespecified analysis. Extended-precision recomputation was not run, so we cannot say whether the smallest retained Superconductivity singular values reflect exact dependence or floating-point rounding. We also did not evaluate how minimum-norm risk changes under alternative pseudoinverse tolerances. The Superconductivity protocol details were finalized after YearPredictionMSD's results but before Superconductivity target labels were accessed, and should be interpreted with that chronological qualification. A relocated Superconductivity manifest retains an obsolete absolute output path; its resolved artifacts match the historical hashes, and the numerical audit detected no disagreement. These limitations narrow inference rather than changing the reported original-scale mean MSEs.

6 Conclusion

Across two distinct fixed-holdout benchmark protocols, source-cross-validated ridge achieves the lowest mean holdout MSE at all 30 dataset-by-size combinations. Minimum-norm regression is severely ill-conditioned around the interpolation threshold, which lies at $n = p + 1$ under training-set centering. On Superconductivity, it suffers rare catastrophic prediction errors that dominate its arithmetic mean and inflate paired uncertainty intervals. Post-hoc diagnostics tie these errors to a few repetitions whose smallest retained singular value sits barely above the pseudoinverse cutoff. The typical repetition is also markedly worse near the threshold, and the mean-risk gap narrows at the largest observed training sizes. Together, these experiments provide reproducible finite-benchmark evidence that spectral regularization can protect prediction risk around the interpolation

threshold, while leaving universal estimator rankings and causal claims about distribution shift unresolved.

References

Madhu S. Advani, Andrew M. Saxe, and Haim Sompolinsky. High-dimensional dynamics of generalization error in neural networks. *Neural Networks*, 132:428–446, 2020. doi: 10.1016/j.neunet.2020.08.022.

Mikhail Belkin, Daniel Hsu, Siyuan Ma, and Soumik Mandal. Reconciling modern machine-learning practice and the classical bias–variance trade-off. *Proceedings of the National Academy of Sciences*, 116(32):15849–15854, 2019. doi: 10.1073/pnas.1903070116.

Thierry Bertin-Mahieux. Year prediction MSD [dataset]. UCI Machine Learning Repository, 2011. URL `https://archive.ics.uci.edu/dataset/203/yearpredictionmsd`.

Edgar Dobriban and Stefan Wager. High-dimensional asymptotics of prediction: Ridge regression and classification. *The Annals of Statistics*, 46(1):247–279, 2018. doi: 10.1214/17-AOS1549.

Kam Hamidieh. A data-driven statistical model for predicting the critical temperature of a superconductor. *Computational Materials Science*, 154:346–354, 2018a. doi: 10.1016/j.commatsci.2018.07.052.

Kam Hamidieh. Superconductivty Data [dataset]. UCI Machine Learning Repository, 2018b. URL `https://archive.ics.uci.edu/dataset/464/superconductivty+data`.

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2 edition, 2009. doi: 10.1007/978-0-387-84858-7.

Trevor Hastie, Andrea Montanari, Saharon Rosset, and Ryan J. Tibshirani. Surprises in high-dimensional ridgeless least squares interpolation. *The Annals of Statistics*, 50(2):949–986, 2022. doi: 10.1214/21-AOS2133.

Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970. doi: 10.1080/00401706.1970.10488634.

William F. Massy. Principal components regression in exploratory statistical research. *Journal of the American Statistical Association*, 60(309):234–256, 1965. doi: 10.1080/01621459.1965.10480787.

Preetum Nakkiran, Prayaag Venkat, Sham M. Kakade, and Tengyu Ma. Optimal regularization can mitigate double descent. In *International Conference on Learning Representations*, 2021.

A Complete mean-risk grids

The primary-endpoint mean MSEs below come from the final comparison table (see Appendix B for provenance). Each row reports the mean over 100 repetitions, and n/p uses $p = 90$ or $p = 81$ as applicable. "Mean only" is the post-hoc training-mean reference. Values are rounded for display; the stored artifacts retain full precision.

Table 4: YearPredictionMSD mean holdout MSE at every sample size.

n	n/p	Minimum-norm	Ridge	PCR	Mean only
45	0.500	422.27	121.73	133.25	121.34
68	0.756	941.24	118.84	129.06	120.18
81	0.900	2,325.90	116.67	126.29	119.54
86	0.956	5,010.31	116.06	125.35	119.26
88	0.978	10,317.86	115.71	125.61	119.25
90	1.000	48,972.31	115.78	123.20	119.21
92	1.022	48,728.59	115.48	123.57	119.20
95	1.056	6,411.13	115.41	124.05	119.14
99	1.100	2,902.68	115.74	124.27	119.16
113	1.256	923.51	114.51	121.70	119.01
135	1.500	465.12	113.17	119.37	118.79
180	2.000	261.17	111.46	119.06	118.53
270	3.000	166.87	108.80	117.54	118.25
450	5.000	125.65	104.51	114.34	118.03
900	10.000	104.68	99.50	105.48	117.92

Table 5: Superconductivity mean holdout MSE at every sample size. Minimum-norm values at $n = 77$–83 are shown to five significant figures because $n = 79$ and $n = 81$ coincide at four (2.6978×10^{17} and 2.6985×10^{17}; their largest-loss repetitions are 57 and 5, respectively).

n	n/p	Minimum-norm	Ridge	PCR	Mean only
41	0.506	8.837×10^{9}	575.29	689.78	1176.02
61	0.753	2.812×10^{12}	530.90	612.30	1165.68
73	0.901	3.238×10^{19}	509.82	610.22	1160.67
77	0.951	2.8370×10^{17}	498.17	593.72	1159.05
79	0.975	2.6978×10^{17}	493.76	589.65	1157.70
81	1.000	2.6985×10^{17}	491.10	594.77	1158.07
83	1.025	2.9270×10^{17}	490.85	591.79	1157.32
85	1.049	5.311×10^{14}	489.39	594.46	1157.52
89	1.099	3.719×10^{5}	483.96	577.34	1157.24
101	1.247	23,590.03	467.75	546.92	1156.37
122	1.506	3,993.00	452.30	520.73	1155.74
162	2.000	1,263.51	419.72	471.56	1152.88
243	3.000	581.58	387.36	428.07	1149.72
405	5.000	407.13	360.10	392.83	1147.83
810	10.000	341.26	333.63	345.19	1146.60

Table 6: Selected paired differences and pointwise 95% Student-t intervals. Positive differences favor ridge for the contrasts shown.

Dataset, n	Contrast	Mean difference	Pointwise 95% interval
Year, 90	PCR − ridge	7.42	[5.21, 9.63]
Year, 90	MN − ridge	48,856.53	[22,299.94, 75,413.11]
Year, 900	MN − ridge	5.18	[4.79, 5.58]
Super, 73	PCR − ridge	100.40	[78.17, 122.63]
Super, 73	MN − ridge	3.238×10^{19}	$[-3.181, 9.657] \times 10^{19}$
Super, 81	MN − ridge	2.698×10^{17}	$[-0.437, 5.834] \times 10^{17}$
Super, 810	PCR − ridge	11.56	[9.68, 13.45]
Super, 810	MN − ridge	7.63	[5.88, 9.38]

B Paired inference, seeds, and provenance

Primary versus descriptive quantities. The main risk comparison uses the arithmetic mean across 100 repetition-level holdout MSEs. For Superconductivity at $n = 73$, the minimum-norm median is 1.396×10^6, the first quartile is 3.093×10^5, and the single largest repetition contributes 0.999169 of the aggregate summed MSE. For YearPredictionMSD at $n = 90$, the minimum-norm median is 15,483.77 and the largest repetition (repetition 7) contributes 0.251 of the summed MSE. Fractions of repetitions in which an estimator "wins" are empirical counts, not additional formal tests.

Seeds and protocol chronology. The training-subset seed lists were stored in the study protocols, with a separate list for each dataset. YearPredictionMSD folds were assigned to 900 maximum-prefix positions using seed 810000001, and Superconductivity folds to 810 positions using seed 820000001. The Superconductivity outer feature-group split uses seed 732111676. The Superconductivity source-pool permutation is applied to the fixed group-permutation concatenation order, not to globally sorted training-row indices. The second protocol was fixed before Superconductivity outcomes were opened, but after YearPredictionMSD results were obtained.

Provenance. YearPredictionMSD run ID: `20261030T085223Z_l2_28abcf33d8e4`. Superconductivity run ID: `20261102T141929Z_super_b3c9fff27517`. A read-only audit reconciled 4,500 estimator-specific metric records per benchmark, the prediction vectors, 45 estimator/size summaries, and 45 paired estimator contrasts. A second audit, run with the post-hoc diagnostics, rechecked the reported win counts, interval signs, and ratio ranges against the per-repetition records and found no discrepancy. The recorded intervals and figures are conditional on each benchmark's fixed holdout.

C Post-hoc diagnostics (not prespecified)

All quantities in this appendix were computed after the primary results were known, using code separate from the confirmatory pipeline and without modifying its outputs.

Table 7: Superconductivity minimum-norm spectral diagnostics by sample size (100 repetitions each). "Repeated rows": prefixes with fewer distinct feature vectors than rows. "Centering only": cells with exactly one truncated value aligned with $\mathbf{1}/\sqrt{n}$ (applicable for $n \le p$). "Near-cutoff": cells with $s_{\min} < 10^{-10} s_1$. "Loss share": fraction of summed minimum-norm MSE in near-cutoff cells. ρ: Pearson correlation of $\log_{10}$ MSE with $\log_{10}(1/s_{\min})$.

n	Repeated rows	Centering only	Max. truncated	Near-cutoff	Loss share (%)	ρ
41	13	87	2	0	0	0.918
61	30	70	3	0	0	0.946
73	39	61	3	3	99.99998	0.975
77	44	56	4	4	99.997	0.976
79	47	53	4	5	99.997	0.976
81	50	50	4	6	99.988	0.975
83	50	—	2	5	99.995	0.969
85	50	—	0	2	99.999	0.957
89	52	—	0	0	0	0.679

Descriptive only: excluding near-cutoff cells, the largest remaining minimum-norm mean is at $n = 81$ (3.5×10^{13}), and the medians are 1.4×10^6, 3.3×10^6, 5.4×10^6, 2.4×10^7, 1.0×10^7, and 1.4×10^6 at $n = 73$, 77, 79, 81, 83, and 85. On YearPredictionMSD, no prefix has repeated rows, all 400 cells with $n \le p$ show the centering-only pattern, no cell is near-cutoff, and ρ ranges from 0.19 ($n = 81$) to 0.64 ($n = 92$).

Table 8: Source-pool correlation-matrix spectra: number of eigenvalues below $c\,\lambda_{\max}$. Both pools have full numerical rank and no duplicated or zero-variance columns.

Dataset	p	$c = 10^{-2}$	10^{-4}	10^{-6}	10^{-8}
YearPredictionMSD	90	4	0	0	0
Superconductivity	81	62	16	0	0

Table 9: Exploratory (not prespecified): Superconductivity minimum-norm versus ridge at the eight sizes whose prespecified mean-difference interval includes zero. Ridge has the lower MSE in 100 of 100 repetitions at each size. The sign test is exact and two-sided; the $\log_{10}$ interval is a pointwise Student-t interval for the mean of $\log_{10} L_{\mathrm{MN}} - \log_{10} L_{\mathrm{Ridge}}$.

n	Sign-test p	Mean $\log_{10}$ difference	95% interval
41	1.6×10^{-30}	1.894	[1.598, 2.190]
61	1.6×10^{-30}	3.419	[2.950, 3.888]
73	1.6×10^{-30}	4.796	[4.149, 5.443]
77	1.6×10^{-30}	5.145	[4.492, 5.798]
79	1.6×10^{-30}	5.356	[4.705, 6.008]
81	1.6×10^{-30}	5.845	[5.176, 6.514]
83	1.6×10^{-30}	5.537	[4.915, 6.160]
85	1.6×10^{-30}	3.881	[3.496, 4.265]

Synthetic illustration of the threshold. With i.i.d. standard Gaussian predictors ($p = 81$), a linear signal, unit noise, the same training-set centering, and the same cutoff τ, every $n \le 81$ truncates exactly one singular value and every $n \ge 82$ truncates none. Over 400 repetitions, the median minimum-norm test MSE is 84.4 at $n = 81$, 212.1 at $n = 82$, and 60.2 at $n = 83$, placing the peak at $n = p + 1$.

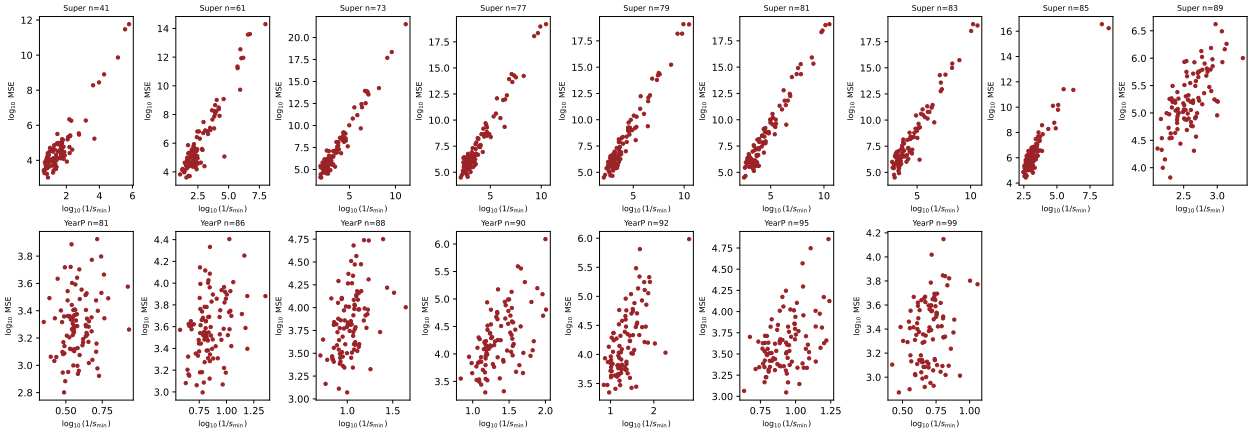


Figure 5: Repetition-level $\log_{10}$ minimum-norm holdout MSE against $\log_{10}(1/s_{\min})$, where $s_{\min}$ is the smallest retained singular value. Top: Superconductivity, $n = 41$–89. Bottom: YearPredictionMSD, $n = 81$–99. Each point is one repetition; axes differ by panel.

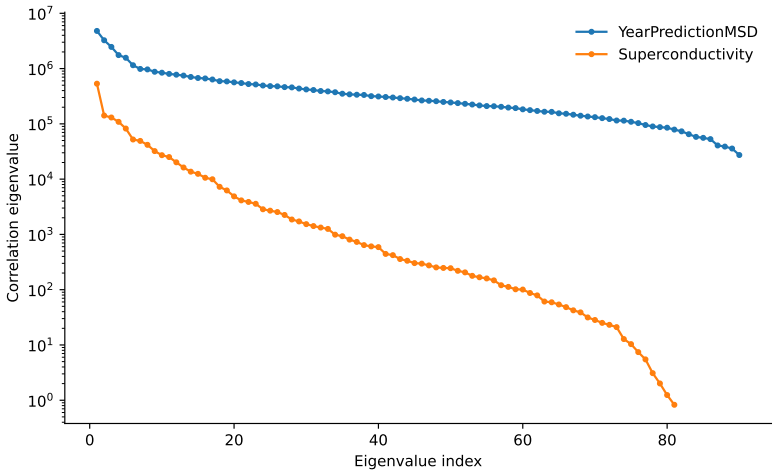


Figure 6: Sorted eigenvalues of the source-pool correlation matrices, on a logarithmic scale.

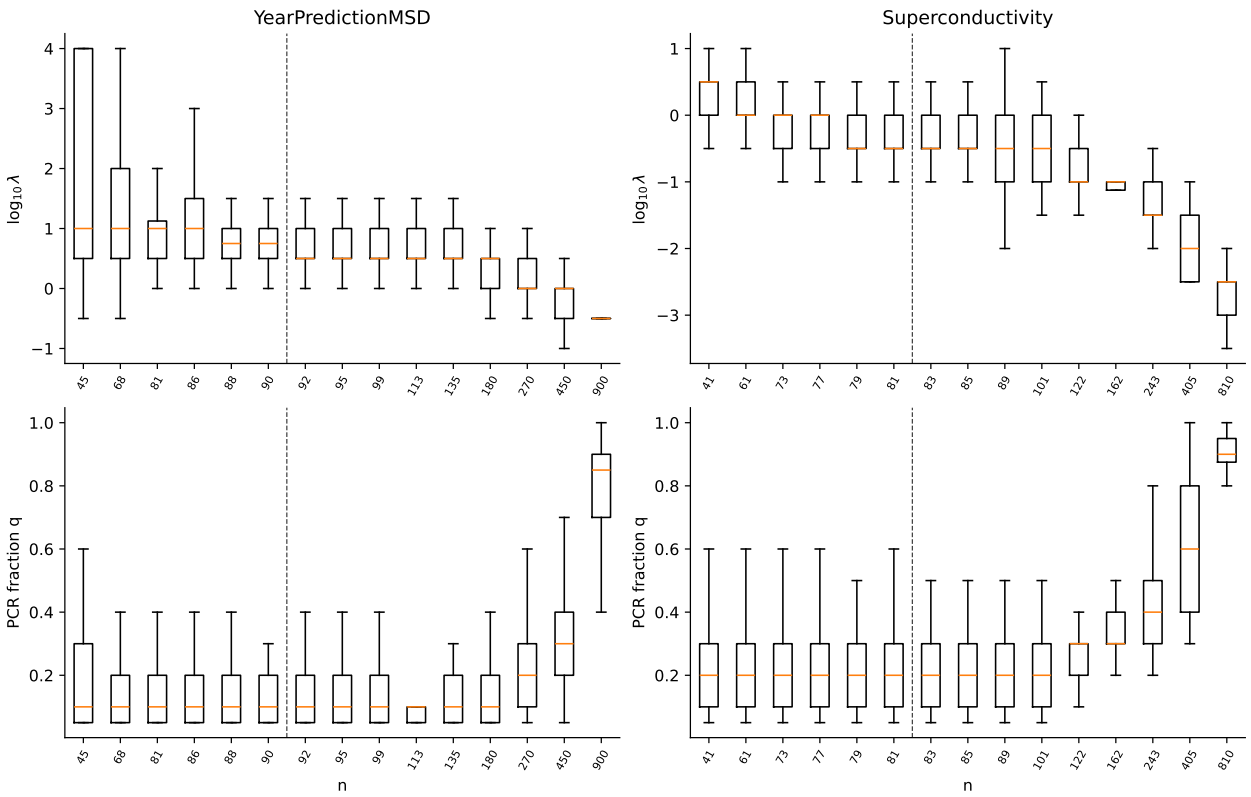


Figure 7: Selected hyperparameters by sample size. Top: $\log_{10}$ of the selected ridge penalty λ (grid 10^{-8} to 10^{4}). Bottom: selected PCR fraction q. The dashed line separates the last grid size with $n \leq p$ from the next size. Boxplot outliers are not drawn.